

Wallace tree

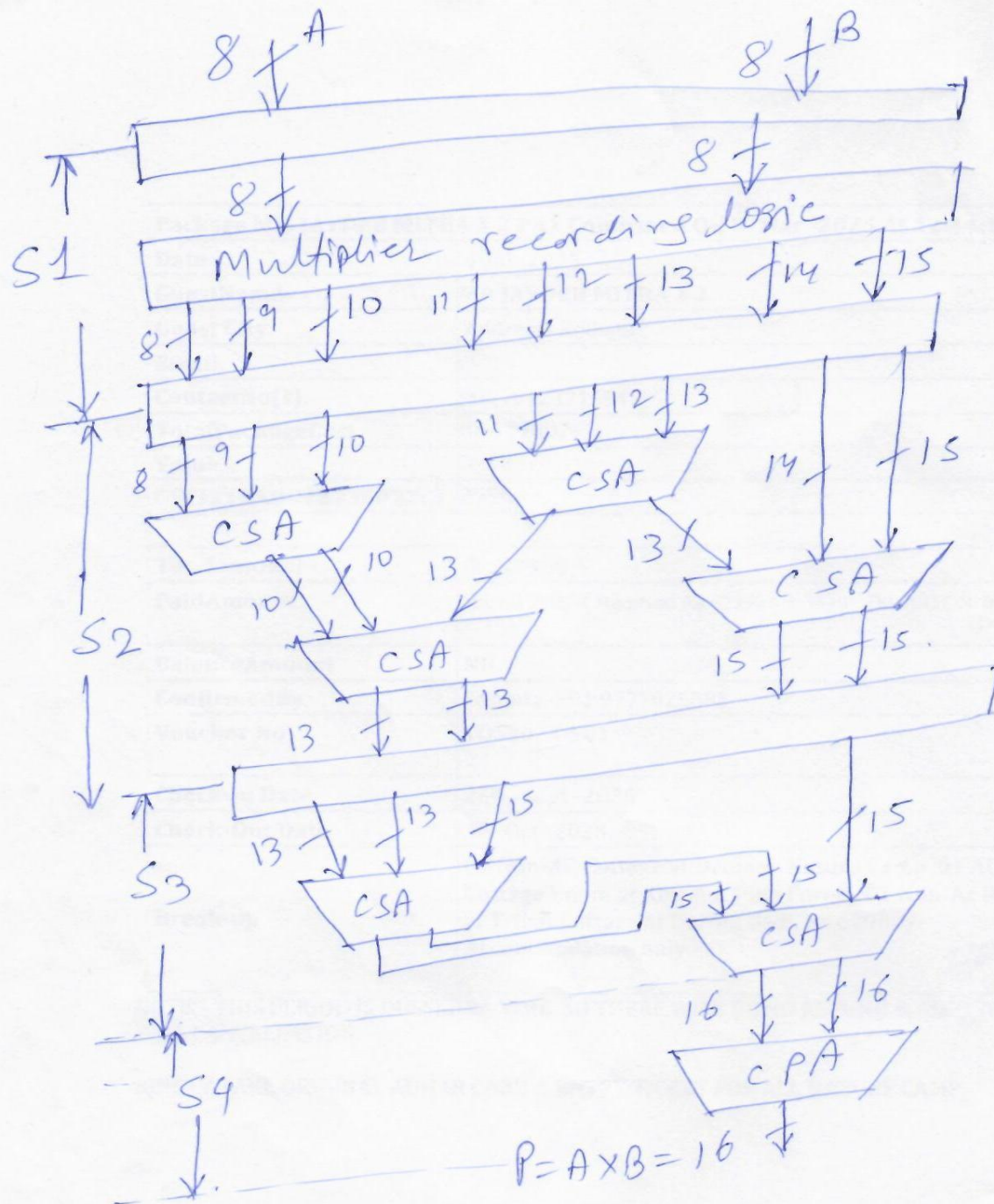
Multiplication of two 8-bit integers $A \times B = P$, where P is the 16-bit product.

$$P = P_0 + P_1 + P_2 + P_3 + P_4 + P_5 + P_6 + P_7$$

[$\times$ is the arithmetic multiply and $+$ is the arithmetic add]

$$\begin{array}{r}
 \begin{array}{r}
 A \quad 1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \\
 B \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 \\
 \hline
 = \quad 1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \\
 \quad 1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \\
 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\
 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\
 \quad 1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \\
 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\
 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\
 \quad 1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\
 \hline
 P \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \\
 =
 \end{array}
 \end{array}$$

Partial product P_j is obtained by multiplying the multiplicand A by the j^{th} bit of the multiplier B and then shifting the result by j bits to the left for $j = 0, 1, 2, 3, 4, 5, 6, 7$. P_j is $(8+j)$ bits long with j trailing zeros. Summation of the eight partial products is done with a Wallace tree of CSAs plus a CPA at the final stage.



S1 generates all 8 partial products, ranging from 8 bits to 15 bits simultaneously

S2, made up of two levels of 4 CSAs, which essentially merges 8 numbers into 4 numbers ranging from 13 to 15 bits

S3 consists of two CSAs, which merges 4 numbers into two 16 bits numbers

S4, final stage to add the last two numbers to produce the final product P

Bit pair multiplication

Booth's algorithm has two attractive features

Useful for positive and negative multipliers uniformly

Improves speed of operation or efficiency by reducing the number of arithmetic operations depending on the multiplier type

Bit pair recording of multiplier reduces the maximum number of operations (partial product) and hence improves speed.

Multiplier bit pair	Multiplier bit on the right	Booth's		Bit pair
$i+1$ i	$i-1$			Multiplicand selected at position i
0 0	0	0 0	$0x2^1 + 0x2^0$	$0 \times M$
0 0	1	0 +1	$0x2^1 + 1x2^0$	$+1 \times M$
0 1	0	+1 -1	$1x2^1 - 1x2^0$	$+1 \times M$
0 1	1	+1 0	$+1x2^1 + 0x2^0$	$+2 \times M$
1 0	0	-1 0	$-1x2^1 + 0x2^0$	$-2 \times M$
1 0	1	-1 +1	$-1x2^1 + 1x2^0$	$-1 \times M$